

RetailHub Sales Analytics System

Company:

RetailHub Pvt. Ltd.

Technology Stack:

- MySQL
- MySQL Workbench
- Git
- GitHub

Company Overview

RetailHub Pvt. Ltd. is a medium-sized Indian retail company selling electronics, home appliances, furniture, fashion, and groceries across multiple physical stores. Its operations encompass a vast network of customers, products, stores, employees, suppliers, orders, and payments. While the company successfully stores data across all these business segments, it currently lacks the capability to extract meaningful business insights.

Current Business Problems

- **Sales Visibility Gaps:** Total absence of tracking for monthly revenue, top products, and seasonal trends.
- **Customer Retention Blind spots:** Inability to identify loyal shoppers, repeat buyer behaviour, or purchase frequency.
- **Inventory Control Failures:** Mismanaged tracking leading to bottlenecks in identifying slow-moving and high-demand products.
- **HR Performance Deficits:** Lack of standardized metrics to measure and compare individual employee sales metrics.
- **Data Silos:** Massive amounts of stored data yield zero actionable operational intelligence for executives.
- **Strategic Blindness:** Leadership cannot determine channel performance, leading to guesswork in scaling or discontinuing products.

Project Objective

The primary objective is to design and implement a structured MySQL analytics database that consolidates RetailHub's operational data. This centralized data warehouse will eliminate current reporting blind spots across sales, inventory, and HR, enabling leadership to build interactive dashboards and extract actionable business insights for data-driven decision-making.

Scope of the Project

This project will include:

- **Relational Schema Design:** Architecting a comprehensive data model mapping customers, products, stores, employees, suppliers, orders, and payments.
- **Database Normalization:** Building optimized MySQL tables utilizing strict primary and foreign key constraints to ensure data integrity.
- **Mock Data Synthesis:** Populating the normalized schema with realistic, clean, and locally relevant datasets for testing.
- **Analytical Query Authoring:** Writing advanced, business-focused SQL scripts to calculate revenue, track seasonality, and identify top performers.
- **Database Object Optimization:** Developing specialized views, stored procedures, automated triggers, and query-speeding indexes.
- **Executive Reporting Insights:** Synthesizing the queried datasets into structured, actionable intelligence ready for leadership dashboards.
- **GitHub Repository Documentation:** Creating an end-to-end technical README file, schema visualisations, and organized .sql script files.

Expected Strategic and Operational Benefits

Here are the strategic and operational benefits expected from implementing the **RetailHub** analytics database:

- **Unified Data Core:** Eliminates fragmented data storage by consolidating sales, customer, inventory, and HR records into a single source of truth.
- **Predictive Inventory Control:** Minimizes capital locked in stagnant stock while preventing costly out-of-stock scenarios for high-demand products.
- **Targeted Customer Marketing:** Identifies high-value, repeat buyers to drive high-ROI retention campaigns and loyalty programs.
- **Optimized Retail Footprint:** Pins down top-performing geographic territories and flags low-performing stores requiring operational interventions.
- **Objective Meritocracy:** Establishes reliable, automated KPIs to fairly reward high-performing sales staff and identify training gaps.
- **Agile Portfolio Management:** Accelerates profitability by isolating declining product categories and pinpointing candidates for discontinuation.
- **Instant Executive Dashboards:** Accelerates management decision-making from days to seconds via clean, pre-optimized views ready for visualization tools.

Project Success Criteria

- Database successfully designed.
- Referential integrity maintained.
- Business queries return accurate results.
- Documentation completed.
- Project uploaded to GitHub.